AGENTIC ECONOMY · ARC HACKATHON

Tessera

Money for machines.

Trustless pay-per-use commerce for AI agents — settled on Arc in USDC. The trust layer that lets an agent pay a service it has never met.

Live on Arc testnet
chainId 5042002 · USDC as gas

8 contracts deployed
escrow, tabs, pool, vault, swap, AMM, 2 collectors

225 automated tests
104 contract · 121 agent · CI on every push

THE PROBLEM

Agents can't pay strangers.

An agent can only buy from a service a human already set up an account and a card for.

Trust runs both ways

The service can't tell the agent will pay. The agent can't tell the service will deliver.

So: pre-funded vendors only

Agents are boxed into a hand-curated list of pre-approved suppliers.

That isn't an economy

Real machine-to-machine commerce needs strangers to transact safely.

THE INSIGHT

The missing primitive isn't “send USDC.” It's escrow + SLA + reputation.

A raw transfer doesn't make a stranger safe to deal with. Programmable escrow — with a delivery guarantee and reputation attached — does.

HOW IT WORKS

402 → decide → escrow → settle or refund

01

Quote

Agent hits a paid endpoint → HTTP 402 plus a signed USDC price and SLA.

02

Decide

Rules or an LLM weigh budget, on-chain reputation, bonded stake and memory.

03

Escrow

Locks a nano-sized USDC payment in TesseraEscrow on Arc.

04

Settle / refund

Delivered → released. SLA breach → reclaimed.

USDC is Arc's gas token, so the agent funds one asset for the toll and the fees — and sub-second finality lets a payment settle inside the request.

WHAT MAKES IT REAL

Guarantees, not promises.

◆ Escrow + auto-refund

Delivery releases funds; an SLA breach reclaims them automatically. Enforced by the contract, not trust.

◆ Provider staking & slashing

Providers bond USDC; a breach slashes it to compensate the agent. Skin in the game.

◆ Nanopayment tabs

One deposit, many zero-gas off-chain vouchers, one settlement. Streams at nano-scale.

◆ Guardian + trust memory

Big buys pause for a human co-signer; a provider that burned this agent is declined next time.

THE DEFI STACK

The rails that fund the agent.

An agent needs working capital. Tessera provides it natively rather than assuming a funded wallet.

◆ Lending & borrowing

Supply for yield or borrow against collateral. Kinked-utilisation interest, health-factor liquidation, per-action freeze controls.

◆ Yield vault

80% held liquid by a contract floor no admin can lower. The app's fee touches yield only — never principal.

◆ Swap desk

Oracle-priced swaps between pool assets, with a configurable fee split.

◆ Liquidity pools (AMM)

Multi-asset pools where providers keep at least 50% of every swap fee — a constant in the contract, not a setting.

GUARDRAILS

What an operator cannot do.

The safety properties that matter are the ones written as constants, not as policy.

Vault reserve floor

80% liquid is a contract constant. Raising it is allowed; lowering it is not.

AMM provider share

50% of swap fees to liquidity providers, enforced on every configuration path.

No position transfers

No function anywhere lets an operator move someone else's position. Migration pays in on their behalf instead.

Oracle validation

A stale, negative, unfinished or carried-over price pauses the market rather than pricing wrongly.

Freeze, not trap

Freezing is per action — withdraw and repay can stay open, and liquidation is never frozen.

AGENT WORKSPACE

Live information the agent can act on.

News across 21 topics, FX, crypto, stocks, indices, commodities, and market analysis derived from those prices.

Named sources

ECB reference rates, CoinGecko, Yahoo Finance, public RSS. Each panel names its source and its age.

Never a fabricated number

An unreachable feed says so. It does not fall back to a stale figure someone might trade on.

Analysis, not opinion

Breadth, leaders, laggards, volatility, dollar direction — arithmetic on the prices shown, no forecasts.

Full transaction history

Filter by user, date, range, value, outcome and type; export to CSV.

LIVE RUN · ONE AUTONOMOUS RUN

AtmosFeed — weather	0.0025	SETTLED
ParityDesk — FX quote	0.0040	SETTLED
WireScoop — news (junk)	0.0030	REFUNDED
AlphaSignal — analysis	0.0080	SETTLED
PulseWire — ticker (tab)	6 ticks	SETTLED

SLA breach → slash

News returns junk; the agent reclaims its USDC and slashes the provider's stake 0.05 → 0.0494.

Guardian pause

A premium call exceeds the policy cap, so a human approves before it settles.

Declined from memory

A later invoice from the provider that failed it is declined, with no human asked.

ON-CHAIN PROOF · ARC TESTNET · CHAINID 5042002

✓ **Settle — USDC released to provider** 0xbbf5c98b…753f
Verified delivery, escrow released

↩ **Refund — SLA breach reclaimed** 0xf8a57c18…18ba
Agent recovered its USDC automatically

⚡ **Tab settle — 6 calls, one transaction** 0x15c13f79…4f66
Nanopayment stream, remainder returned

Escrow 0x9776498c…5a9a7d   Tab 0x5d482c70…d78c6d   chainId 5042002 · live

WHY ARC + CIRCLE

Built for money that moves itself.

USDC is the gas token

The agent funds one asset for both the toll and the fees — no native-gas bootstrap for a bot.

Sub-second finality

A per-call payment clears inside the request/response cycle.

Config, not fork

Chain id, RPC, explorer and USDC address are env-driven; mainnet is a configuration change.

EXECUTION & QUALITY

Every feature runs end to end.

225	**CI**	**1**	**Live**
automated tests 104 contract · 121 agent	full agentic scenario on every push	command to deploy live bootstrap:arc	deployed & transacting on Arc testnet

THE VISION

The settlement rail for agent-to-agent commerce.

Escrow, refunds, staking and reputation that make autonomous commerce safe. Next: real providers, a discovery registry, reputation staking.

github.com/rgdtnhat/arc · live on Arc testnet